

Initial Project Planning Template

Date	26 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Environment Setup & Data Acquisition	USN-01	Set up Python 3.8+ environment, Anaconda Navigator, Jupyter Notebook, and install libraries (Pandas, NumPy, Scikit-Learn, XGBoost, Flask).	2	High	Sathi Krishna Ganeswara Reddy	25-Jun-2026	29-Jun-2026
Sprint-1	Environment Setup & Data Acquisition	USN-02	Download historical credit card applicant dataset, audit features (Gender, Income Type, Education, Credit Score, etc.), and assess relationships.	2	High	Gatreddi Hariprasad	25-Jun-2026	29-Jun-2026

Sprint-1	Exploratory Data Analysis (EDA)	USN-03	Generate comprehensive visual data insights using Matplotlib and Seaborn via Count Plots and Distribution Plots.	3	Medium	Miriyala Makara Jyothi	25-Jun-2026	29-Jun-2026
Sprint-2	Data Preprocessing & Feature Engineering	USN-04	Clean raw dataset by handling missing values, removing duplicate entries, and optimizing scale consistency.	3	High	Tumu Durga Manjusha Krishna Veni	25-Jun-2026	29-Jun-2026
Sprint-2	Data Preprocessing & Feature Engineering	USN-05	Encode categorical string indicators into analytical numerical format and map multi-class payment states to binary targets.	4	High	Vasanth Vardhan Polumati	25-Jun-2026	29-Jun-2026
Sprint-3	Model Building & Evaluation	USN-06	Develop baseline Machine Learning classification architectures including Logistic Regression and Decision Tree Classifiers.	3	Medium	Sathi Krishna Ganeswara Reddy	30-Jun-2026	03-Jul-2026
Sprint-3	Model Building & Evaluation	USN-07	Train Random Forest and optimized XGBoost Classifiers, evaluating performance metrics (Accuracy, Confusion Matrix, Classification Report) to select the champion model.	5	High	Gatreddi Hariprasad	30-Jun-2026	03-Jul-2026
Sprint-3	Model Building & Evaluation	USN-08	Serialize the best performing champion XGBoost model (91.28% accuracy) into a portable binary file format (.pkl).	1	High	Miriyala Makara Jyothi	30-Jun-2026	03-Jul-2026
Sprint-4	Web Interface & Cloud Integration	USN-09	Initialize core Flask application architecture backend (app.py) and design the main prediction intake form interface (index.html).	5	High	Tumu Durga Manjusha Krishna Veni	30-Jun-2026	03-Jul-2026

Sprint-4	Web Interface & Cloud Integration	USN-10	Construct conditional result templates (result_approved.html / result_rejected.html) and model metrics analysis charts dashboard (performance.html).	4	High	Vasanth Vardhan Polumati	30-Jun-2026	03-Jul-2026
Sprint-4	Web Interface & Cloud Integration	USN-11	Deploy model workspace runtime infrastructure to the IBM Watson Machine Learning cloud environment and conduct end-to-end user acceptance testing.	3	Medium	All Members	30-Jun-2026	03-Jul-2026